Talent & Learning Analytics - Overview

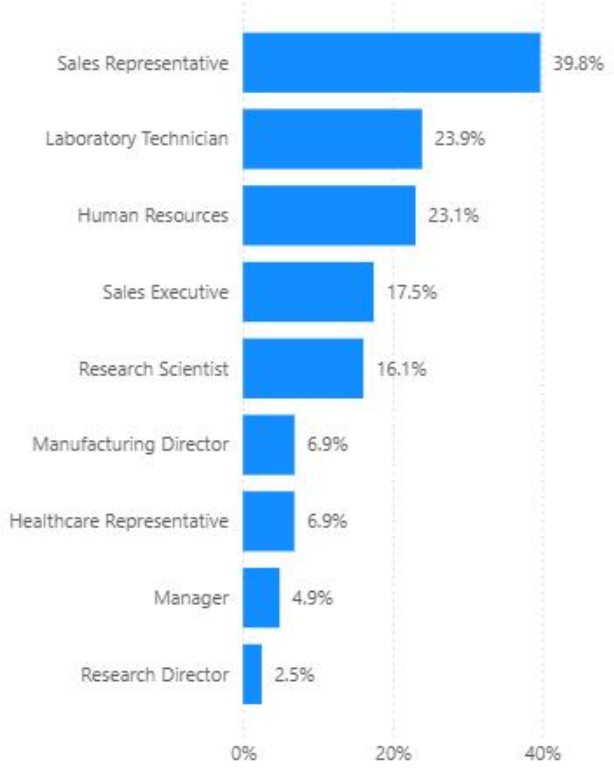
1,470 employees (IBM HR sample data, fictional) - point-in-time snapshot. Use the slicers to filter; right-click a job role to drill through.

Department: All

Job role: All

1,470	237	16.1%	7.0	2.80	28.3%
Headcount	Leavers	Attrition Rate	Avg Tenure (Years)	Training per Employee	Overtime %

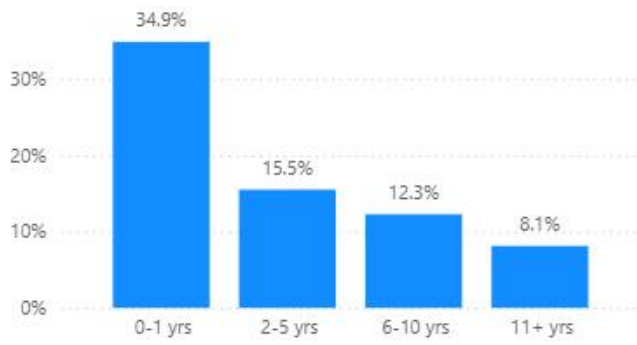
Attrition rate by job role



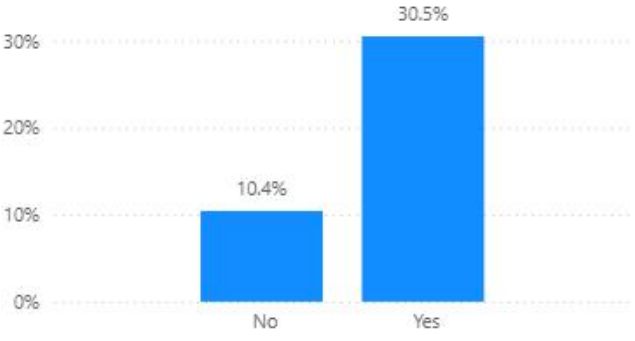
Attrition rate by department



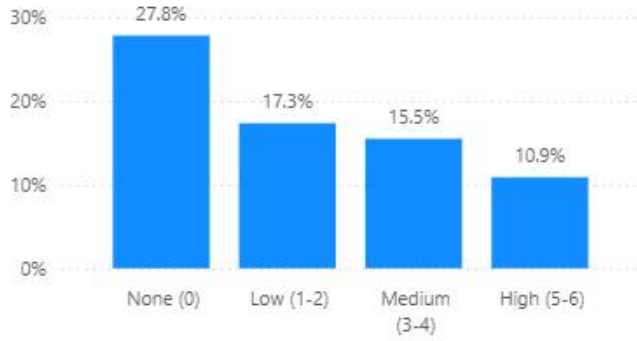
Attrition rate by tenure (years at company)



Attrition rate: overtime vs no overtime



Attrition rate by trainings last year



Insights & Actions

Four evidence-based actions for HR and L&D. Cards are live; details in 04_Docs/Insights_and_Recommendations.md.

2.9x	34.9%	22.8%	16.1%
Overtime Attrition Ratio	Early Tenure Attrition Rate (0-1 yrs)	Attrition Rate - Low Job Satisfaction	Company Attrition Rate

1. Reduce sustained overtime

Attrition 30.5% with overtime vs 10.4% without; holds after controlling for role, tenure and pay.
Action: monthly overtime reviews, rebalance workload. Start with Sales Reps, Lab Technicians, HR, Research Scientists.

2. Make the first year stick

34.9% of 0-1 year employees left (55.1% with overtime). New joiners get no extra training.
Action: 90-day onboarding, mentor per new hire, front-loaded learning path, no routine overtime in year 1.

3. Retention plan for Sales Representatives

39.8% attrition (+23.6 pts); 91.6% are level 1, avg income 2,626 vs 6,503.
Action: pay and incentive benchmark, career path to Sales Executive, coaching.

4. Act on engagement signals

Low job satisfaction: 22.8% attrition; bad work-life balance: 31.3%.
Action: quarterly pulse survey, manager follow-up within 30 days, flexible work options.

Associations in a fictional snapshot (IBM sample), not proven causes - test actions with a pilot first. Also improve learning data: course type, hours, completion, dates. Full analysis: 04_Docs/Insights_and_Recommendations.md

Attrition Drivers

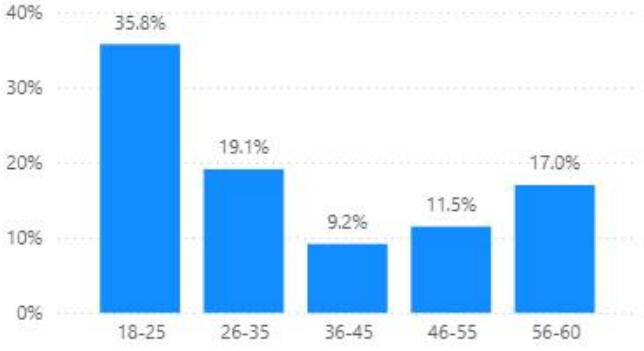
Which groups leave most? Compare each bar with the company rate of 16.1%.

Department: All

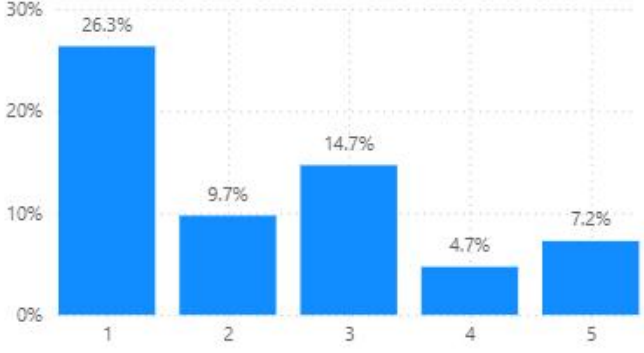
Job role: All

30.5%	10.4%	2.9x	34.9%	22.8%
Attrition Rate - Overtime	Attrition Rate - No Overtime	Overtime Attrition Ratio	Early Tenure Attrition Rate (0-1 yrs)	Attrition Rate - Low Job Satisfaction

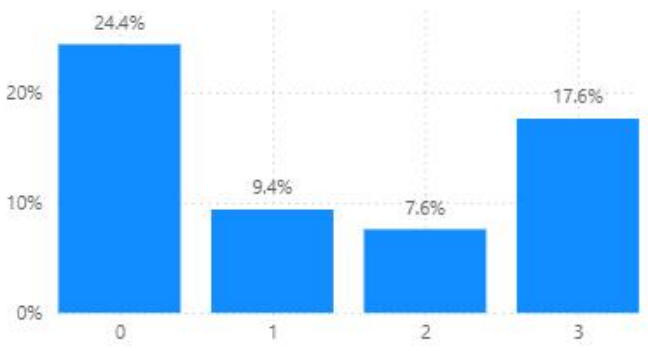
Attrition rate by age band



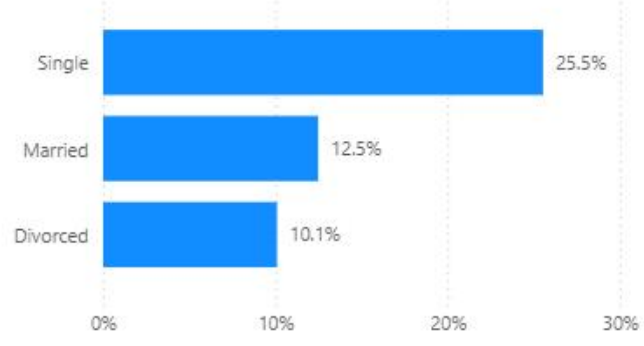
Attrition rate by job level (1 = junior)



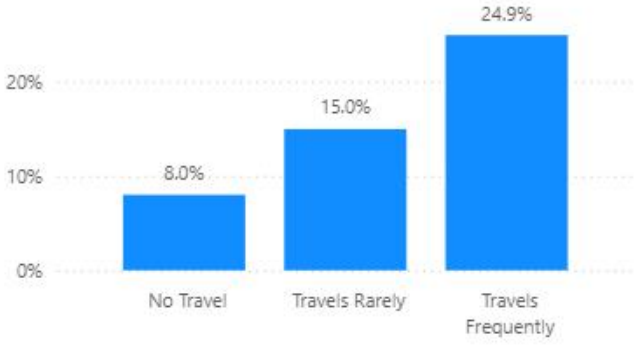
Attrition rate by stock option level



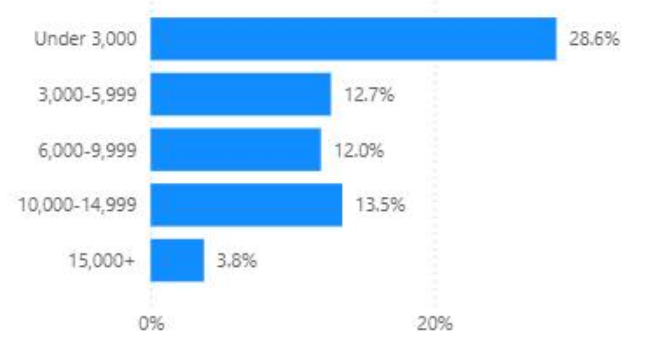
Attrition rate by marital status



Attrition rate by business travel



Attrition rate by monthly income band



Training & Learning

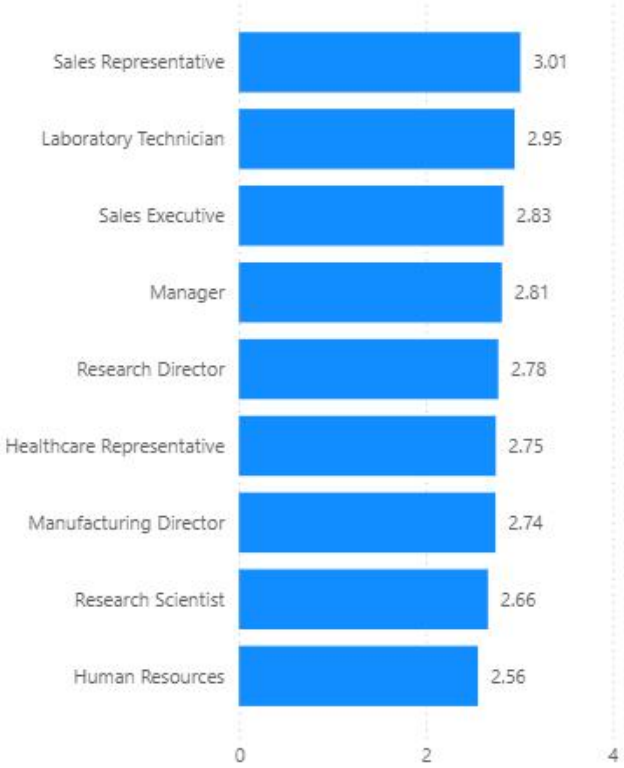
Trainings attended last year (0-6). Note: only 54 employees had no training - treat that group as indicative.

Department: All

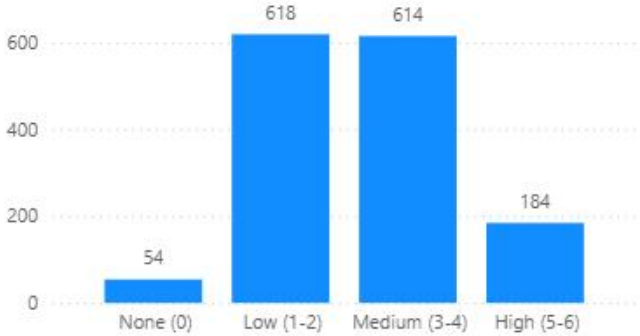
Job role: All

4,115	2.80	54	3.7%	12.5%
Total Trainings	Training per Employee	Employees with No Training	% No Training	% High Training (5-6)

Trainings per employee by job role



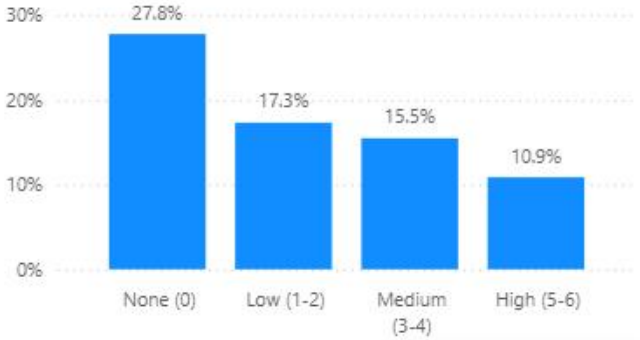
Employees by training band



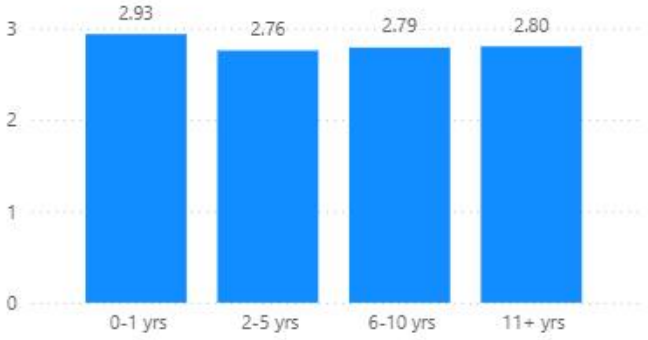
Trainings per employee by department



Attrition rate by training band



Trainings per employee by tenure



Satisfaction & Performance

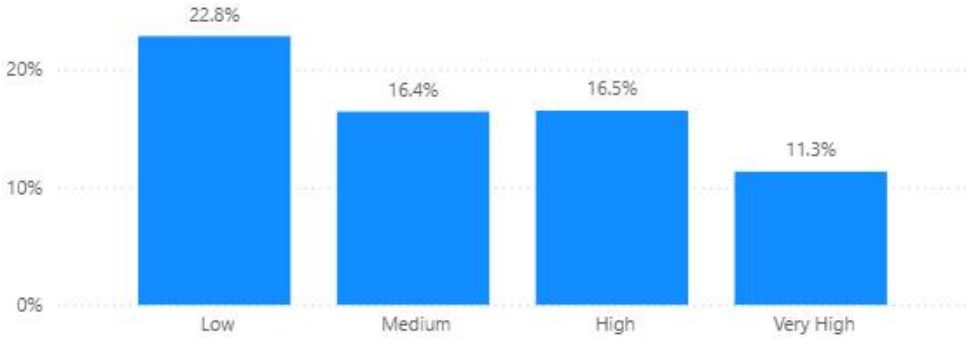
Ratings 1-4. Low job satisfaction and bad work-life balance go with higher attrition; performance rating does not.

Department: All

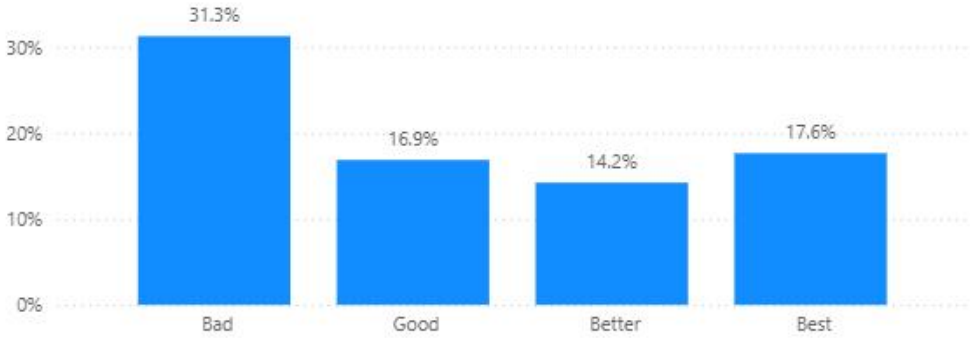
Job role: All

2.73	19.7%	22.8%	2.76	3.15
Avg Job Satisfaction	% Low Job Satisfaction	Attrition Rate - Low Job Satisfaction	Avg Work-Life Balance	Avg Performance Rating

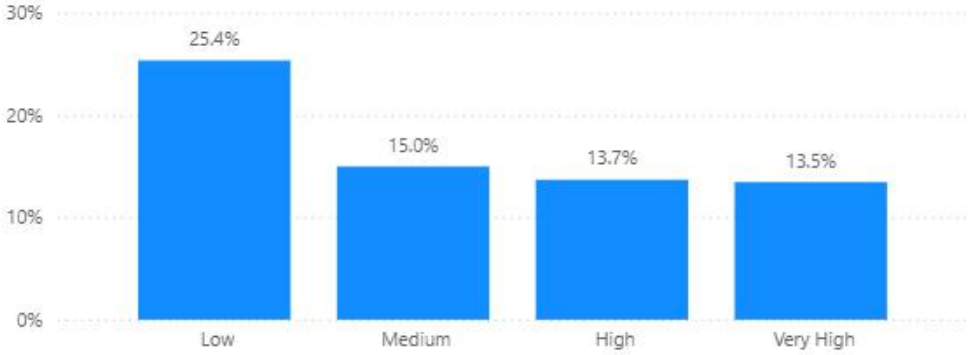
Attrition rate by job satisfaction



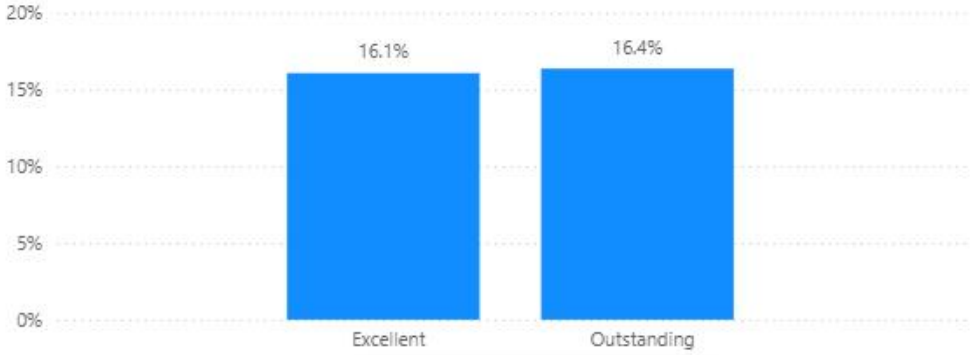
Attrition rate by work-life balance



Attrition rate by environment satisfaction



Attrition rate by performance rating



Job Role Summary

All key measures per job role. Right-click a job role and choose Drill through > Employee Details to see its employees.

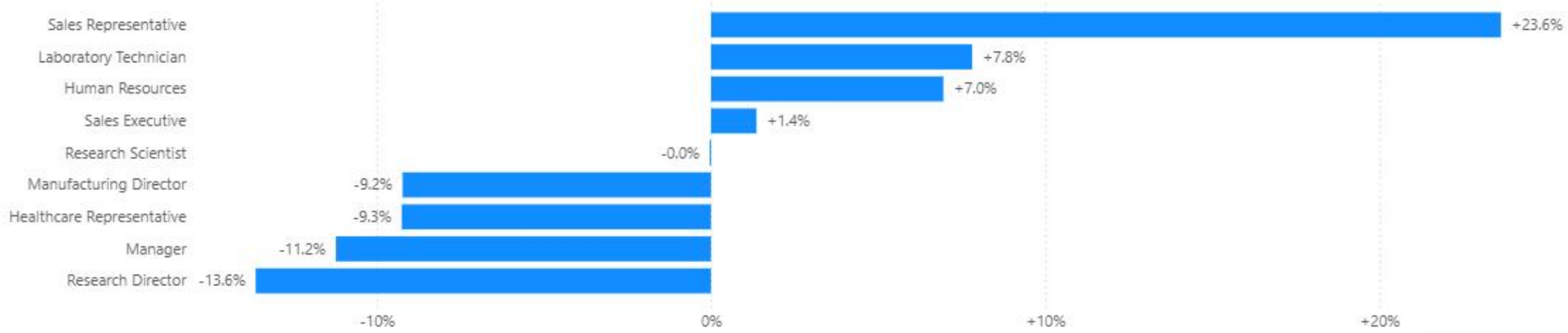
Department: All

Job role: All

Key measures by job role (sorted by attrition gap vs company)

Job role	Headcount	Leavers	Attrition Rate	Attrition vs Company (pts)	Avg Tenure (Years)	Training per Employee	% No Training	Overtime %	Avg Job Satisfaction
Sales Representative	83	33	39.8%	+23.6%	2.9	3.01	1.2%	28.9%	2.73
Laboratory Technician	259	62	23.9%	+7.8%	5.0	2.95	2.3%	23.9%	2.69
Human Resources	52	12	23.1%	+7.0%	5.3	2.56	3.8%	25.0%	2.56
Sales Executive	326	57	17.5%	+1.4%	7.5	2.83	3.4%	28.8%	2.75
Research Scientist	292	47	16.1%	-0.0%	5.1	2.66	5.8%	33.2%	2.77
Manufacturing Director	145	10	6.9%	-9.2%	7.6	2.74	4.1%	26.9%	2.68
Healthcare Representative	131	9	6.9%	-9.3%	8.4	2.75	4.6%	28.2%	2.79
Manager	102	5	4.9%	-11.2%	14.4	2.81	2.0%	26.5%	2.71
Research Director	80	2	2.5%	-13.6%	10.9	2.78	3.8%	28.8%	2.70
Total	**1,470**	**237**	**16.1%**	**0.0%**	**7.0**	**2.80**	**3.7%**	**28.3%**	**2.73**

Attrition gap vs company rate, in percentage points (positive = more leavers than average)



Employee Details

Drill-through page: right-click a job role on any page > Drill through > Employee Details. Opened directly it shows all employees.

1,470	237	16.1%	2.80
Headcount	Leavers	Attrition Rate	Training per Employee

Employees in the current selection

Employee ID	Status	Department	Job role	Gender	Overtime	Age	Tenure (years)	Trainings last year	Job satisfaction (1-4)	Monthly income
1	Left	Sales	Sales Executive	Female	Yes	41	6	0	4	5,993
2	Active	Research & Development	Research Scientist	Male	No	49	10	3	2	5,130
4	Left	Research & Development	Laboratory Technician	Male	Yes	37	0	3	3	2,090
5	Active	Research & Development	Research Scientist	Female	Yes	33	8	3	3	2,909
7	Active	Research & Development	Laboratory Technician	Male	No	27	2	3	2	3,468
8	Active	Research & Development	Laboratory Technician	Male	No	32	7	2	4	3,068
10	Active	Research & Development	Laboratory Technician	Female	Yes	59	1	3	1	2,670
11	Active	Research & Development	Laboratory Technician	Male	No	30	1	2	3	2,693
12	Active	Research & Development	Manufacturing Director	Male	No	38	9	2	3	9,526
13	Active	Research & Development	Healthcare Representative	Male	No	36	7	3	3	5,237
14	Active	Research & Development	Laboratory Technician	Male	No	35	5	5	2	2,426
15	Active	Research & Development	Laboratory Technician	Female	Yes	29	9	3	3	4,193
16	Active	Research & Development	Research Scientist	Male	No	31	5	1	3	2,911
18	Active	Research & Development	Laboratory Technician	Male	No	34	2	2	4	2,661
19	Left	Research & Development	Laboratory Technician	Male	Yes	28	4	4	3	2,028
20	Active	Research & Development	Manufacturing Director	Female	No	29	10	1	1	9,980
21	Active	Research & Development	Research Scientist	Male	Yes	32	6	5	2	3,298
22	Active	Research & Development	Laboratory Technician	Male	Yes	22	1	2	4	2,935
23	Active	Sales	Manager	Female	No	53	25	3	4	15,427
24	Active	Research & Development	Research Scientist	Male	Yes	38	3	3	4	3,944
26	Active	Research & Development	Manufacturing Director	Female	No	24	4	5	3	4,011
27	Left	Sales	Sales Representative	Male	No	36	5	4	1	3,407

Data Quality (self-testing model)

35 automatic checks recalculated on every refresh: row counts, duplicates, nulls, orphan keys and control totals. All must be PASS.

QA_Checks - recalculated on every refresh

#	Check	Actual	Expected	Result
1	FactEmployee rows	1,470	1,470	PASS
2	DimEmployee rows	1,470	1,470	PASS
3	FactEmployee columns	28	28	PASS
4	DimEmployee columns	20	20	PASS
5	DimDepartment rows	3	3	PASS
6	DimJobRole rows	9	9	PASS
7	DimEducation rows	5	5	PASS
8	DimJobSatisfaction rows	4	4	PASS
9	DimAgeBand rows	5	5	PASS
10	DimTenureBand rows	4	4	PASS
11	DimIncomeBand rows	5	5	PASS
12	DimTrainingBand rows	4	4	PASS
13	Duplicate EmployeeID in FactEmployee	0	0	PASS
14	Duplicate EmployeeID in DimEmployee	0	0	PASS
15	Duplicate DepartmentKey in DimDepartment	0	0	PASS
16	Duplicate JobRoleKey in DimJobRole	0	0	PASS
17	Rows with a null in FactEmployee	0	0	PASS
18	Rows with a null in DimEmployee	0	0	PASS
19	Orphan EmployeeID (Fact -> DimEmployee)	0	0	PASS
20	Orphan DepartmentKey	0	0	PASS
21	Orphan JobRoleKey	0	0	PASS
22	Orphan EducationKey	0	0	PASS
23	Orphan JobSatisfactionKey	0	0	PASS
24	Orphan AgeBandKey	0	0	PASS
25	Orphan TenureBandKey	0	0	PASS
26	Orphan IncomeBandKey	0	0	PASS
27	Orphan TrainingBandKey	0	0	PASS